```
import pandas as pd
import numpy as np

np.random.seed(42)

n = 2000

# Generate data
data = {
    "patient_id": range(1, n+1),

    # Age: realistic hospital range with some noise
    "age": np.random.randint(18, 90, n),

    # BMI: normal + some extreme values for cleaning
    "bmi": np.round(np.random.normal(27, 6, n), 1),

    # Gender
    "gender": np.random.choice(["Male", "Female"], n),

    # Conditions
    "hypertension": np.random.choice([0, 1], n, p=[0.7, 0.3]),
    "diabetes": np.random.choice([0, 1], n, p=[0.75, 0.25]),
    "heart_disease": np.random.choice([0, 1], n, p=[0.85, 0.15]),

    # Health metrics
    "avg_glucose_level": np.round(np.random.normal(120, 40, n), 1),

    # Lifestyle
    "smoking_status": np.random.choice(
        ["never", "former", "smokes"], n, p=[0.6, 0.25, 0.15]
    ),
}

df = pd.DataFrame(data)

# --- Create realistic target (IMPORTANT) ---
# Higher risk if older + diseases + high BMI + high glucose
risk_score = (
    (df["age"] > 60).astype(int) +
    df["hypertension"] +
    df["diabetes"] +
    df["heart_disease"] +
    (df["bmi"] > 30).astype(int) +
    (df["avg_glucose_level"] > 150).astype(int)
)

# Convert to probability
prob = 1 / (1 + np.exp(-risk_score))

df["readmitted"] = np.random.binomial(1, prob)

# --- Introduce REAL-WORLD MESSINESS (VERY IMPORTANT FOR EASY PART) ---
# Missing values
df.loc[np.random.choice(n, 100), "bmi"] = np.nan
df.loc[np.random.choice(n, 50), "smoking_status"] = None

# Invalid values
df.loc[np.random.choice(n, 20), "age"] = -5   # wrong age
df.loc[np.random.choice(n, 20), "bmi"] = 80   # extreme BMI

# Save
df.to_csv("hospital_records.csv", index=False)

print("Dataset created successfully with shape:", df.shape)
```

```
Dataset created successfully with shape: (2000, 10)
```

EASY Sub-step 1: Data Quality Audit

```
import pandas as pd

df = pd.read_csv('/content/hospital_records.csv')

print(df.shape)
df.head()
```

```
(2000, 10)
```

	patient_id	age	bmi	gender	hypertension	diabetes	heart_disease	avg_glucose_level	smoking_status	readmitted
0	1	69	21.0	Male	0	0	0	88.3	never	0
1	2	32	26.8	Female	1	0	0	154.9	never	1
2	3	89	15.2	Female	0	0	1	90.5	smokes	1
3	4	78	39.4	Female	0	0	0	42.0	former	1
4	5	38	NaN	Male	1	0	0	147.9	former	1

Next steps: Generate code with df New interactive sheet

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 10 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   patient_id         2000 non-null   int64
 1   age                2000 non-null   int64
 2   bmi                1904 non-null   float64
 3   gender             2000 non-null   object
 4   hypertension       2000 non-null   int64
 5   diabetes           2000 non-null   int64
 6   heart_disease      2000 non-null   int64
 7   avg_glucose_level  2000 non-null   float64
 8   smoking_status     1950 non-null   object
 9   readmitted         2000 non-null   int64
dtypes: float64(2), int64(6), object(2)
memory usage: 156.4+ KB
```

	patient_id	age	bmi	hypertension	diabetes	heart_disease	avg_glucose_level	readmitted
count	2000.000000	2000.000000	1904.000000	2000.000000	2000.000000	2000.000000	2000.00000	2000.00000
mean	1000.500000	52.129000	27.457773	0.296500	0.217000	0.152000	118.78490	0.78700
std	577.494589	21.457343	8.031800	0.456828	0.412306	0.359111	39.29228	0.40953
min	1.000000	-5.000000	6.200000	0.000000	0.000000	0.000000	-18.20000	0.00000
25%	500.750000	34.000000	22.800000	0.000000	0.000000	0.000000	92.07500	1.00000
50%	1000.500000	52.000000	27.000000	0.000000	0.000000	0.000000	118.15000	1.00000
75%	1500.250000	71.000000	31.300000	1.000000	0.000000	0.000000	144.65000	1.00000
max	2000.000000	89.000000	80.000000	1.000000	1.000000	1.000000	261.80000	1.00000

```
df.isnull().sum()
```

	0
patient_id	0
age	0
bmi	96
gender	0
hypertension	0
diabetes	0
heart_disease	0
avg_glucose_level	0
smoking_status	50
readmitted	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
for col in df.columns:
    print(col, df[col].unique())
```

```
patient_id [   1    2    3 ... 1998 1999 2000]
age [69 32 89 78 38 41 20 39 70 19 47 55 81 77 50 75 66 76 59 79 64 68 72 24
 56 35 21 31 26 88 61 25 52 53 67 23 71 80 51 65 57 43 58 46 62 82 18 28
 22 45 29 40 54 44 60 30 49 83 74 87 73 -5 36 85 34 86 33 37 84 48 42 63
 27]
bmi [21.  26.8 15.2 39.4  nan 21.8 13.2 24.3 13.3 28.7 28.2 33.9 20.3 17.4
 24.6 26.5 32.5 35.5 34.3 31.8 23.2 28.  19.9 27.4 27.5 37.3 25.6 28.4
 22.2 34.8 21.1 19.  31.1 22.8 22.3 32.3 30.6 34.9 19.5 24.  21.4 30.9
 23.5 27.2 80.  37.8 31.5 21.9 29.2 30.3 25.7 23.4 19.3 16.6 15.3 30.5
 27.9 24.1 25.  28.3 26.  28.8 28.6 32.  22.1 26.2 31.4 20.8 30.  29.
 26.4 31.9 18.  31.3 40.8 33.5 15.6 37.9 30.7 32.4 23.3 29.9 25.5 26.9
 34.7 24.5 18.6 35.7 18.3 20.4 14.4 16.8 37.1 22.5 34.  23.6 28.1 22.
 40.3 25.2 27.8 22.9 38.2 38.3 36.8 32.7 27.  33.7 23.1 10.4 24.4 21.7
 18.8 20.5 24.9 34.2 15.4 29.1 18.5 20.  23.8 31.6 33.2 25.8 24.8 29.4
 35.3 26.7 24.7  9.8 31.7 21.6 21.5 36.4 27.6 20.2 16.4 31.2 26.3 30.4
 41.5 25.1 23.7 30.8 33.1 43.  29.3 19.4 39.1 37.6 22.7 35.4 26.6 23.9
 22.4 30.1 28.9 22.6 21.2 16.  35.2 34.1 30.2 19.7 27.7 32.9 25.4 35.6
 13.6 33.  23.  14.8 34.5 17.9 34.4 19.1 31.  33.6 28.5 25.9 21.3 29.5
 15.7 33.3 29.6 24.2 43.1 16.1 35.8 39.6 20.9 12.6 35.9 40.7 13.8 14.6
 29.7 19.6 43.4 14.  18.7 15.9 36.2 13.7 32.1 29.8 38.  36.5 27.3 26.1
 17.2 34.6 20.1 16.2 17.8 18.1 20.6 32.8 41.  38.8 33.8 32.2 12.5 43.3
 38.7 25.3 27.1 17.1 40.1 38.9 20.7 33.4 10.  18.4 36.6 36.3 16.5 11.2
 18.9 13.4 37.  19.2 17.3 17.6 17.5 42.8  6.2 17.  11.6 16.7 13.1 32.6
 12.7 38.6 36.1 37.2 36.  36.7 39.8 37.5 14.1 14.2 35.1 13.  39.2 14.5
 18.2 39.7 19.8 35.  45.4 15.5  7.1 17.7 39.5 38.5 15.8 13.9 16.3 41.1
 11.  13.5 15.1 12.8 14.3 38.4 37.4 43.6 42.3 40.  39.  41.7]
gender ['Male' 'Female']
hypertension [0 1]
diabetes [0 1]
heart_disease [0 1]
avg_glucose_level [ 88.3 154.9  90.5 ...  98.8 162.5  87.8]
smoking_status ['never' 'smokes' 'former' nan]
readmitted [0 1]
```

```
# Age issues
print(df[df['age'] < 0])

# BMI issues
print(df[df['bmi'] > 60])

# Glucose issues
print(df[df['avg_glucose_level'] < 0])
```

```
1439               79.5         smokes           1
1474              102.4         former           0
1680              108.2          never           1
1682              100.6            NaN           1
1754              164.7          never           1
1894               87.7         former           1
      patient_id  age   bmi  gender  hypertension  diabetes  heart_disease  \
56            57   71  80.0  Female             1         0              0
116          117   32  80.0    Male             0         0              0
193          194   88  80.0    Male             0         1              1
```

```
1779              191.2          never           1
1795              161.3          never           1
1921              138.5          never           1
1992              129.9         smokes           1
      patient_id  age   bmi  gender  hypertension  diabetes  heart_disease  \
18            19   39  26.5    Male             0         0              0
846          847   23  29.8  Female             1         0              0
1878        1879   33  27.7  Female             0         0              0

      avg_glucose_level smoking_status  readmitted
18                 -5.7            NaN           0
846                -0.7          never           0
1878              -18.2          never           0
```

EASY Sub-step 2: Data Cleaning

```
# Fill BMI with median
df['bmi'].fillna(df['bmi'].median(), inplace=True)

# Fill smoking_status with mode
df['smoking_status'].fillna(df['smoking_status'].mode()[0], inplace=True)
```

```
/tmp/ipykernel_23267/4275218902.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['bmi'].fillna(df['bmi'].median(), inplace=True)
/tmp/ipykernel_23267/4275218902.py:5: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['smoking_status'].fillna(df['smoking_status'].mode()[0], inplace=True)
```

```
# Remove negative age
df = df[df['age'] >= 0]

# Cap BMI
df['bmi'] = df['bmi'].clip(10, 50)
```

```
# Cap glucose
df['avg_glucose_level'] = df['avg_glucose_level'].clip(50, 300)
```

```
# Gender encoding
df['gender'] = df['gender'].map({'Male': 0, 'Female': 1})

# Smoking encoding
df = pd.get_dummies(df, columns=['smoking_status'], drop_first=True)
```

```
df.isnull().sum()
```

	0
patient_id	0
age	0
bmi	0
gender	0
hypertension	0
diabetes	0
heart_disease	0
avg_glucose_level	0
readmitted	0
smoking_status_never	0
smoking_status_smokes	0

dtype: int64

```
df.head()
```

	patient_id	age	bmi	gender	hypertension	diabetes	heart_disease	avg_glucose_level	readmitted	smoking_status_never
0	1	69	21.0	0	0	0	0	88.3	0	True
1	2	32	26.8	1	1	0	0	154.9	1	True
2	3	89	15.2	1	0	0	1	90.5	1	False
3	4	78	39.4	1	0	0	0	50.0	1	False
4	5	38	27.0	0	1	0	0	147.9	1	False

Next steps: Generate code with df New interactive sheet

MEDIUM Sub-step 3: Build Neural Network

```
import numpy as np

# Drop ID column
df_model = df.drop(columns=['patient_id'])

# Convert boolean columns to int for numerical operations
for col in ['smoking_status_never', 'smoking_status_smokes']:
    if col in df_model.columns:
        df_model[col] = df_model[col].astype(int)

# Split features and target
X = df_model.drop(columns=['readmitted']).values
y = df_model['readmitted'].values.reshape(-1, 1)

# Normalize features (IMPORTANT)
X = (X - X.mean(axis=0)) / X.std(axis=0)

# Train-test split
split = int(0.8 * len(X))
X_train, X_test = X[:split], X[split:]
y_train, y_test = y[:split], y[split:]
```

```
input_size = X_train.shape[1]
hidden_size = 16
output_size = 1

np.random.seed(42)

W1 = np.random.randn(input_size, hidden_size) * 0.01
b1 = np.zeros((1, hidden_size))

W2 = np.random.randn(hidden_size, output_size) * 0.01
b2 = np.zeros((1, output_size))
```

```
def sigmoid(x):
    return 1 / (1 + np.exp(-x))

def sigmoid_derivative(x):
    return x * (1 - x)
```

```
def forward(X):
    Z1 = np.dot(X, W1) + b1
    A1 = sigmoid(Z1)

    Z2 = np.dot(A1, W2) + b2
    A2 = sigmoid(Z2)

    return A1, A2
```

```
def compute_loss(y, y_hat):
    m = y.shape[0]
    loss = - (1/m) * np.sum(
        y * np.log(y_hat + 1e-8) + (1 - y) * np.log(1 - y_hat + 1e-8)
    )
    return loss
```

```
def backward(X, y, A1, A2, lr=0.01):
    global W1, b1, W2, b2
    m = X.shape[0]

    dZ2 = A2 - y
```

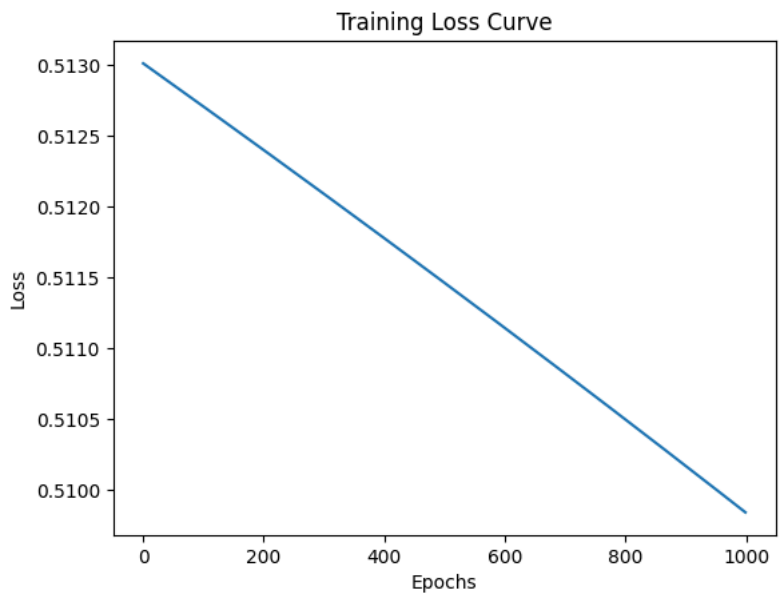
```
    dW2 = (1/m) * np.dot(A1.T, dZ2)
    db2 = (1/m) * np.sum(dZ2, axis=0, keepdims=True)

    dA1 = np.dot(dZ2, W2.T)
    dZ1 = dA1 * sigmoid_derivative(A1)
    dW1 = (1/m) * np.dot(X.T, dZ1)
    db1 = (1/m) * np.sum(dZ1, axis=0, keepdims=True)

    # Update
    W1 -= lr * dW1
    b1 -= lr * db1
    W2 -= lr * dW2
    b2 -= lr * db2
```

MEDIUM Sub-step 4: Train + Evaluate

```
losses = []

for epoch in range(1000):
    A1, A2 = forward(X_train)
    loss = compute_loss(y_train, A2)
    losses.append(loss)

    backward(X_train, y_train, A1, A2, lr=0.01)

    if epoch % 100 == 0:
        print(f"Epoch {epoch}, Loss: {loss:.4f}")
```

```
Epoch 0, Loss: 0.5130
Epoch 100, Loss: 0.5127
Epoch 200, Loss: 0.5124
Epoch 300, Loss: 0.5121
Epoch 400, Loss: 0.5118
Epoch 500, Loss: 0.5115
Epoch 600, Loss: 0.5111
Epoch 700, Loss: 0.5108
Epoch 800, Loss: 0.5105
Epoch 900, Loss: 0.5102
```

```
import matplotlib.pyplot as plt

plt.plot(losses)
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.title("Training Loss Curve")
plt.show()
```




```
# Predict on the test set using the custom neural network
_, y_pred_prob = forward(X_test)
y_pred_nn_custom = (y_pred_prob > 0.5).astype(int)
```

```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        83
           1       0.79      1.00      0.88       314

    accuracy                           0.79       397
   macro avg       0.40      0.50      0.44       397
weighted avg       0.63      0.79      0.70       397

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-de
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-de
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-de
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train.ravel())

y_pred_sklearn = model.predict(X_test)

print(classification_report(y_test, y_pred_sklearn))
```

```
              precision    recall  f1-score   support

           0       0.58      0.13      0.22        83
           1       0.81      0.97      0.88       314

    accuracy                           0.80       397
   macro avg       0.69      0.55      0.55       397
weighted avg       0.76      0.80      0.74       397
```

MEDIUM Sub-step 5: Business Decision

```
cost_fn = 10   # missing patient (FN)
cost_fp = 1    # false alarm (FP)
```

```
thresholds = np.arange(0.1, 0.9, 0.1)

for t in thresholds:
    preds = (y_pred_prob > t).astype(int)

    TP = np.sum((preds == 1) & (y_test == 1))
    TN = np.sum((preds == 0) & (y_test == 0))
    FP = np.sum((preds == 1) & (y_test == 0))
    FN = np.sum((preds == 0) & (y_test == 1))

    cost = FN * cost_fn + FP * cost_fp

    print(f"Threshold {t:.1f} → Cost: {cost}")
```

```
Threshold 0.1 → Cost: 83
Threshold 0.2 → Cost: 83
Threshold 0.3 → Cost: 83
Threshold 0.4 → Cost: 83
Threshold 0.5 → Cost: 83
Threshold 0.6 → Cost: 83
Threshold 0.7 → Cost: 83
Threshold 0.8 → Cost: 2231
```

HARD Sub-step 6: Why 94% Accuracy is Misleading

```
# Check class distribution
print(df['readmitted'].value_counts())
```

```
readmitted
1    1560
0     421
Name: count, dtype: int64
```

```
from sklearn.linear_model import LogisticRegression

X = df.drop(columns=['patient_id', 'readmitted'])
y = df['readmitted']

model = LogisticRegression()
model.fit(X, y)
```

```
y_pred = model.predict(X)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

```
from sklearn.metrics import accuracy_score

print("Accuracy:", accuracy_score(y, y_pred))
```

```
Accuracy: 0.7874810701665825
```

```
from sklearn.metrics import confusion_matrix

print(confusion_matrix(y, y_pred))
```

```
[[  25  396]
 [  25 1535]]
```

```
from sklearn.metrics import classification_report

print(classification_report(y, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.50      0.06      0.11       421
           1       0.79      0.98      0.88      1560

    accuracy                           0.79      1981
   macro avg       0.65      0.52      0.49      1981
weighted avg       0.73      0.79      0.72      1981
```

HARD Sub-step 7: Neural Network as Feature Extractor

```
df_nn_features = df.drop(columns=['patient_id'])

for col in ['smoking_status_never', 'smoking_status_smokes']:
    if col in df_nn_features.columns:
        df_nn_features[col] = df_nn_features[col].astype(int)

X_for_forward = df_nn_features.drop(columns=['readmitted']).values
X_for_forward = (X_for_forward - X_for_forward.mean(axis=0)) / X_for_forward.std(axis=0)


A1, _ = forward(X_for_forward)

# A1 = hidden layer output (embeddings)
X_embed = A1
```

```
from sklearn.linear_model import LogisticRegression

clf = LogisticRegression()
clf.fit(X_embed[:split], y_train.ravel())

y_pred_embed = clf.predict(X_embed[split:])
```

```
print("Custom Neural Network:")
print(classification_report(y_test, y_pred_nn_custom))

print("Logistic Regression with NN Embeddings:")
print(classification_report(y_test, y_pred_embed))
```

```
Custom Neural Network:
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        83
           1       0.79      1.00      0.88       314

    accuracy                           0.79       397
   macro avg       0.40      0.50      0.44       397
```

```
weighted avg       0.63      0.79      0.70       397

Logistic Regression with NN Embeddings:
              precision    recall  f1-score   support

           0       1.00      0.01      0.02        83
           1       0.79      1.00      0.88       314

    accuracy                           0.79       397
   macro avg       0.90      0.51      0.45       397
weighted avg       0.84      0.79      0.70       397

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-de
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-de
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-de
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

Start coding or generate with AI.

Start coding or generate with AI.